

Integers and Absolute Value

Lesson 9-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*... -3, -2, -1, 0, 1, 2, 3 ... shown on a number line
with 0 in the center*

Integer

A whole number that can be positive, negative, or zero.

*+5 means 5 above zero, like 5 floors above ground
level*

Positive

A number bigger than zero, to the right of zero on a number line.

-3 means 3 below zero, like 3 floors underground

Negative

A number smaller than zero, to the left of zero on a number line.

$|-5| = 5$ and $|5| = 5$ — both are 5 units from 0

Absolute value

How far a number is from zero. It is never negative.

3 and -3 are opposites — both are 3 units from 0

Opposite

Two numbers the same distance from zero, on opposite sides, like 3 and -3.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can identify integers and find their absolute value.

1 Notice

Captain Vega's treasure map has clues hidden at different elevations!

2 Key idea

I can identify integers and find their absolute value.

3 Words I'll use

Integer

Positive

Negative

Absolute value

Opposite



Think About It

- What does the negative sign tell you about the treasure chest's location?
- Which location is farther from sea level — the chest at -4 or the tower at $+6$?
- How can you find the distance from sea level without worrying about direction?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Why are -5 and 5 both the same distance from zero on the number line, even though one is negative and one is positive?

Level 1 support

Start here: -5 and 5 are the same distance from zero because ____.

 **Need a hint?**

Sentence starters (optional)

Both numbers are ____ units from zero.

Ambos numeros estan a ____ unidades del cero.

They are different because one is ____ and one is ____.

Son diferentes porque uno es ____ y el otro es ____.

Word bank: integer absolute value opposite positive negative

Listen for: Student says both are 5 units from zero (same absolute value) but are opposites because of their signs.

Level 2

Push students to explain why absolute value can never be negative.

- Absolute value is never negative because ____.
- The absolute value of any number tells me ____.

EXPLORE

A diver is at -30 feet and a hawk is at +30 feet. How can absolute value help you describe how far each is from sea level?

Level 1 support

Start here: Absolute value tells me how far each animal is from ____.



Need a hint?

Sentence starters (optional)

The diver is ____ feet from sea level because $|-30| =$ ____.

El buzo esta a ____ pies del nivel del mar porque $|-30| =$ ____.

The hawk is ____ feet from sea level because $|30| =$ ____.

El halcon esta a ____ pies del nivel del mar porque $|30| =$ ____.

Word bank: **integer** **absolute value** **positive** **negative** **opposite**

Listen for: Student uses absolute value to show both are 30 feet from sea level (zero), ignoring direction.

Level 2

Ask students to decide who is truly farther from sea level and justify using both the integers and their absolute values.

- They are the same distance from sea level because ____.
- The sign tells me the direction, while the absolute value tells me ____.

CONNECT

How is the opposite of a number different from the absolute value of a number?

Level 1 support

Start here: The opposite changes ____, but the absolute value gives ____.



Need a hint?

Sentence starters (optional)

The opposite of 7 is ____.

El opuesto de 7 es ____.

The absolute value of -7 is ____.

El valor absoluto de -7 es ____.

Word bank: integer absolute value opposite positive negative

Listen for: Student explains the opposite flips the sign while absolute value always gives the non-negative distance from zero.

Level 2

Push students to find a number whose opposite and absolute value are equal, and explain why.

- The opposite and absolute value are equal when the number is ____.
- This works because ____.

REFLECT

Think of a real situation where a negative integer makes sense. What does zero mean in that situation?

Level 1 support

Start here: A negative integer makes sense when ____, and zero means ____.



Need a hint?

Sentence starters (optional)

A negative number could describe ____.

Un numero negativo podria describir ____.

In that situation, zero means ____.

En esa situacion, el cero significa ____.

Word bank: integer negative positive absolute value opposite

Listen for: Student gives a real context (temperature, elevation, money) and explains zero as the reference point between positive and negative.

Level 2

Ask students to critique whether absolute value would be useful in their chosen situation and why.

- Absolute value would help in my situation because ____.
- It would tell me ____ no matter the direction.

Guided Examples

Example 1

What is the absolute value of -9?

Solution: Absolute value is the distance from zero. -9 is 9 units from zero, so $|-9| = 9$.

Answer: A. 9

Example 2

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

Solution: $|-3| = 3$, $|7| = 7$, $|-5| = 5$, $|2| = 2$. The greatest absolute value is 7.

Answer: A. 7

Example 3

What is $|0|$?

Solution: Zero is 0 units from itself on the number line, so $|0| = 0$.

Answer: A. 0

Write About the Math

The Writing Revolution

I can explain my reasoning using the words integer, positive, negative, absolute value, and opposite.

1. Kernel Sentence subject + verb

Model: Integer is a whole number that can be positive, negative, or zero.

Número entero es un número entero que puede ser positivo, negativo o cero.

Write a kernel sentence about integer. Use a subject and a verb.

Escribe una oración base sobre número entero. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Integer matters in math

Número entero importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Integer matters in math because ____.

Número entero importa en matemáticas porque ____.

but
pero

Integer matters in math, but ____.

Número entero importa en matemáticas, pero ____.

so
entonces

Integer matters in math, so ____.

Número entero importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about integer.

Di un hecho verdadero sobre integer.

Integer ____.

Question *Pregunta*

Ask a question about integer.

Haz una pregunta sobre integer.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about integer.

Muestra entusiasmo sobre integer.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with integer.

Dile a un compañero qué hacer con integer.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Both numbers are ____ units from zero.

Ambos numeros estan a ____ unidades del cero.

They are different because one is ____ and one is ____.

Son diferentes porque uno es ____ y el otro es ____.

The diver is ____ feet from sea level because $|-30| =$ ____.

El buzo esta a ____ pies del nivel del mar porque $|-30| =$ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $|0|$?

- A. 0
- B. 1
- C. -1
- D. There is no answer

Show your work:

2. Which statement about absolute value is TRUE?

- A. $|-4| = |4|$ because both are 4 units from zero
- B. $|-4| = -4$ because the number is negative
- C. $|4| > |-4|$ because positive is always bigger
- D. $|-4| = 0$ because negatives cancel out

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A submarine is at -150 meters and a bird is flying at +150 meters. A student says the submarine is closer to sea level because -150 is less than 150. Is the student correct? Explain using absolute value.

Sentence starter: The student is ____ because absolute value measures _____. Since $|-150| = \underline{\hspace{1cm}}$ and $|150| = \underline{\hspace{1cm}}$, both are ____ from sea level. The student confused ____ with ____.

Show your work:

Reflect — Exit Ticket

Which integer has an absolute value of 8 and is located to the LEFT of zero on the number line?

- A. 8
- B. -8
- C. 0
- D. -80

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 0 — *Zero is 0 units from itself on the number line, so $|0| = 0$.*
2. **Try It 2:** A. $|-4| = |4|$ because both are 4 units from zero — *Absolute value measures distance from zero. Both -4 and 4 are 4 units from zero, so $|-4| = |4| = 4$.*
3. **Exit Ticket:** B. -8 — *To the left of zero means negative. The integer -8 has an absolute value of 8 because it is 8 units from zero.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Integer is a whole number that can be positive, negative, or zero.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).